

BLACK DIAMOND, AB, Canada

WMO: 719820

Lat: 50.7067N

Lon: 114.1517W

Elev: 1156

StdP: 88.19

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-29.3	-26.1	-32.4	0.2	-29.1	-29.3	0.3	-25.9	12.9	-0.4	11.4	-0.2	2.1	250	0.684

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.8	28.2	15.2	26.2	14.8	24.3	14.1	17.2	24.0	16.3	23.4	15.3	22.3	4.2	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.6	11.9	20.6	13.5	11.1	19.4	12.5	10.4	18.0	53.0	24.3	50.0	23.7	46.9	22.3	20.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.6	8.9	7.7	-33.6	31.8	3.0	2.7	-35.8	33.7	-37.5	35.3	-39.2	36.8	-41.4	38.7		
(5)			-33.8	19.3	2.9	0.8	-35.9	19.9	-37.6	20.4	-39.3	20.8	-41.4	21.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.4	-6.7	-8.3	-2.6	3.2	8.8	12.5	15.4	14.9	10.4	4.0	-3.0	-8.6		
	DBStd	10.47	8.01	8.26	7.77	4.80	4.16	2.95	2.66	3.03	4.62	4.96	7.13	8.09		
	HDD10.0	2920	518	512	390	208	73	9	1	2	49	191	389	577		
	HDD18.3	5458	777	746	648	454	296	176	96	112	238	443	639	836		
	CDD10.0	517	0	0	1	3	37	85	168	154	62	6	1	0		
	CDD18.3	13	0	0	0	0	0	1	6	5	1	0	0	0		
	CDH23.3	635	0	0	0	4	30	51	241	231	79	0	0	0		
	CDH26.7	147	0	0	0	0	2	8	60	62	14	0	0	0		
(14) Wind	WSAvg	3.2	3.1	2.8	3.4	3.6	3.6	3.5	3.1	2.8	3.0	3.1	3.2	2.9		
(15) Precipitation	PrecAvg	598	28	25	35	48	78	107	62	62	52	34	34	29		
	PrecMax	932	51	39	89	92	145	379	160	132	146	60	81	64		
	PrecMin	407	10	4	15	28	20	17	16	11	5	5	3	10		
	PrecStd	110	10	9	17	18	36	69	38	29	32	16	18	14		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.0	11.9	15.5	23.1	26.5	28.0	30.7	31.3	28.6	21.6	17.5	10.0		
		MCWB	4.3	5.5	6.2	9.8	12.2	16.2	15.6	15.0	13.5	10.4	7.5	3.9		
	2%	DB	8.1	7.9	12.9	19.0	23.8	24.6	28.3	28.0	25.5	18.3	12.6	7.1		
		MCWB	2.6	2.5	5.1	8.3	11.3	13.9	16.0	14.8	13.4	9.1	5.8	2.2		
	5%	DB	6.3	5.4	10.5	15.5	21.4	22.5	26.0	25.7	23.1	15.8	9.3	4.9		
		MCWB	1.9	0.7	3.8	6.3	10.5	13.0	15.7	14.7	12.5	7.9	3.8	0.5		
	10%	DB	4.4	3.0	7.8	12.4	18.8	20.8	23.8	23.4	20.2	13.2	6.6	2.5		
		MCWB	0.6	-0.5	2.6	4.6	9.3	12.3	15.3	14.6	11.3	6.5	2.1	-0.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.9	5.7	7.4	10.1	13.1	17.4	18.8	17.9	15.3	11.3	8.0	4.3		
		MCDB	10.5	11.2	13.8	22.2	23.1	24.4	25.3	24.4	24.7	19.4	15.6	9.4		
	2%	WB	3.1	2.6	5.5	8.3	11.9	15.4	17.6	16.7	14.2	9.8	6.3	2.3		
		MCDB	7.5	7.3	12.0	17.8	22.3	21.9	24.2	24.0	22.9	17.1	12.4	6.9		
	5%	WB	2.0	1.1	4.1	6.7	10.9	14.0	16.7	15.9	13.0	8.2	4.1	0.7		
		MCDB	6.1	4.9	9.9	14.5	20.1	20.8	23.6	23.1	21.5	15.1	8.9	4.2		
	10%	WB	0.5	-0.6	2.7	5.1	9.8	13.0	15.8	14.9	11.8	6.7	2.4	-0.7		
		MCDB	4.3	2.6	7.7	10.9	17.7	19.1	22.8	21.8	19.3	12.7	6.1	2.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.6	12.0	12.2	13.0	13.7	13.3	14.8	14.8	14.6	13.5	12.3	11.5		
		MCDBR	13.3	14.5	15.7	19.2	19.1	17.2	19.9	20.0	20.4	18.0	15.5	13.2		
		MCWBR	9.1	9.9	9.3	9.8	8.6	8.4	9.2	8.7	9.4	9.7	9.4	9.2		
		MCDBR	13.1	14.0	14.9	18.7	17.3	15.3	16.2	16.0	18.4	17.4	15.3	13.3		
	5% WB	MCWBR	9.1	9.9	9.2	9.9	8.1	8.1	8.9	8.2	8.7	9.6	9.5	9.5		
(40) Clear-Sky Solar Irradiance	taub		0.251	0.272	0.281	0.337	0.342	0.344	0.340	0.356	0.313	0.278	0.254	0.236		
	taud		2.437	2.419	2.465	2.313	2.372	2.406	2.425	2.384	2.485	2.560	2.511	2.440		
	Ebn at Noon		813	879	935	904	913	912	909	876	886	867	807	780		
	Edh at Noon		57	76	88	116	114	112	108	107	86	66	53	49		
(44) All-Sky Solar Radiation	RadAvg		1.14	2.11	3.31	4.64	5.43	5.82	6.51	5.34	3.75	2.32	1.23	0.89		
	RadStd		0.08	0.16	0.24	0.32	0.43	0.44	0.43	0.37	0.40	0.24	0.10	0.06		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (70 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page